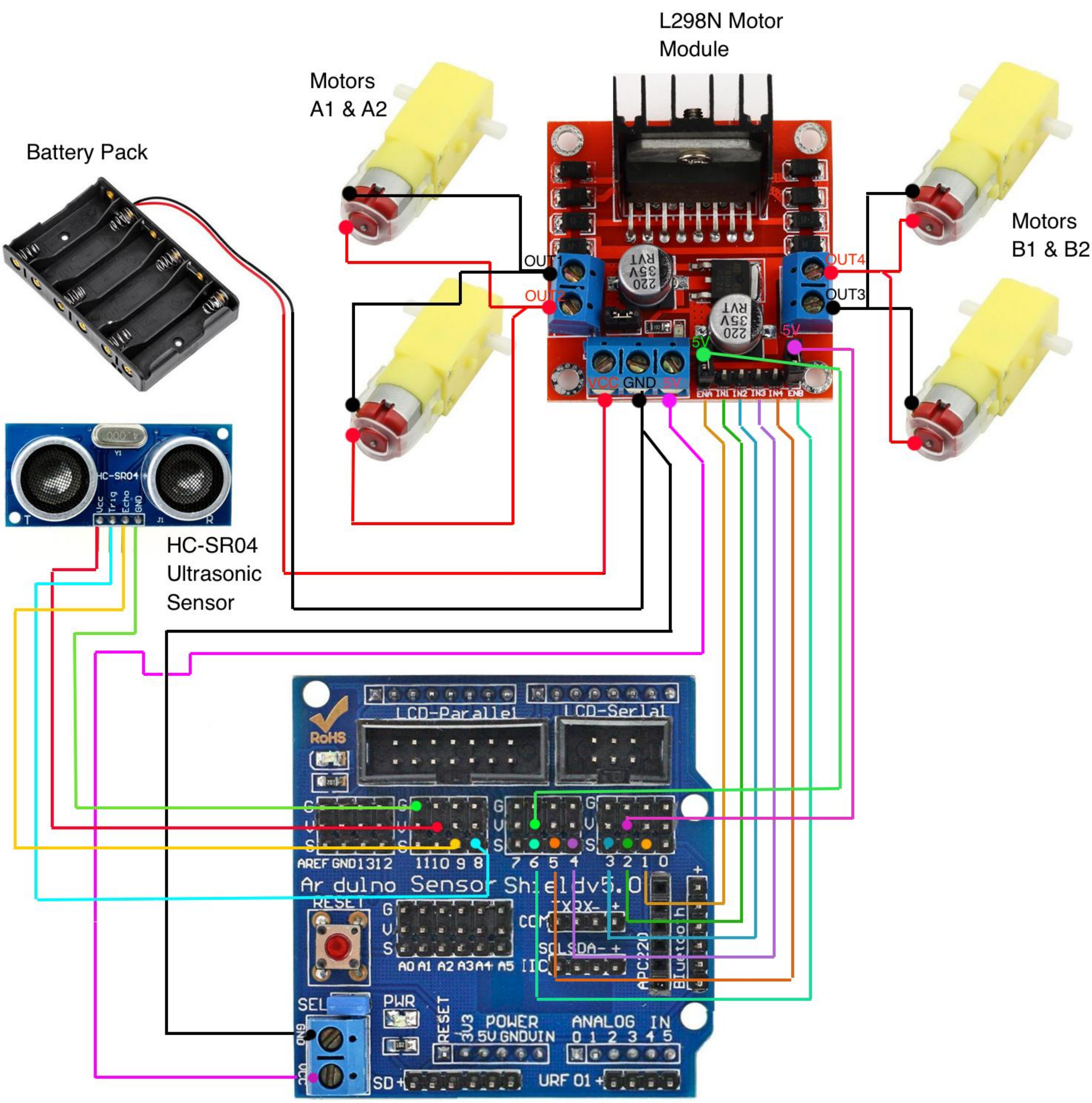
Hardware Assembly and Wiring
The robot car was assembled using a 4WD chassis, four DC motors, an Arduino Uno, an L298N motor driver, and an HC-SR04 ultrasonic sensor.
Motor Driver Connections
- The positive terminal of the battery pack was connected to the VCC pin of the L298N motor driver.
- The negative terminal of the battery pack was connected to the GND pin of the motor driver.
- The two left-side motors were connected to OUT1 and OUT2.
- The two right-side motors were connected to OUT3 and OUT4.

Arduino Connections
- ENA was connected to Arduino digital pin 1.
- IN1 was connected to digital pin 2.
- IN2 was connected to digital pin 3.
- IN3 was connected to digital pin 4.
- IN4 was connected to digital pin 5.
- ENB was connected to digital pin 6.

Ultrasonic Sensor Connections
The HC-SR04 ultrasonic sensor was mounted at the front of the robot to detect obstacles.
- TRIG → Digital Pin 8
- ECHO → Digital Pin 9
- VCC → 5V
- GND → GND

Power Supply
The Arduino Uno was powered using a 9V battery connected through the DC barrel jack, while the motors were powered by the external battery pack connected to the motor driver.
After completing all wiring connections, the obstacle avoidance code was uploaded to the Arduino board. The robot moves forward continuously and automatically changes direction whenever an obstacle is detected within 15 cm.



Arduino Sensor Shield v5.0